State Spaces, Algebras, and Flows

Algebraic solutions of Projects 9.1 and 9.2

Heze Zhang

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Main presentation

1. State space and algebra

2. Project 9.1: affine dynamics

3. Project 9.2: general flows

Appendix: derivations and supplementary results

Questions

- How does the algebra of trajectories solve an affine ODE?
- Which flow laws remain valid for nonlinear dynamics?

Detailed arguments are linked from the main slides.

Assessment, Project 9, items 1–2. Manuscript §§ 1.1–1.2 and Appendix A.

State space and algebra

Vector space

$$V = \mathbb{R}^d, \qquad x = (x_1, \ldots, x_d).$$

Addition and scalar multiplication:

$$(x + y)_i = x_i + y_i,$$
$$(\lambda x)_i = \lambda x_i.$$

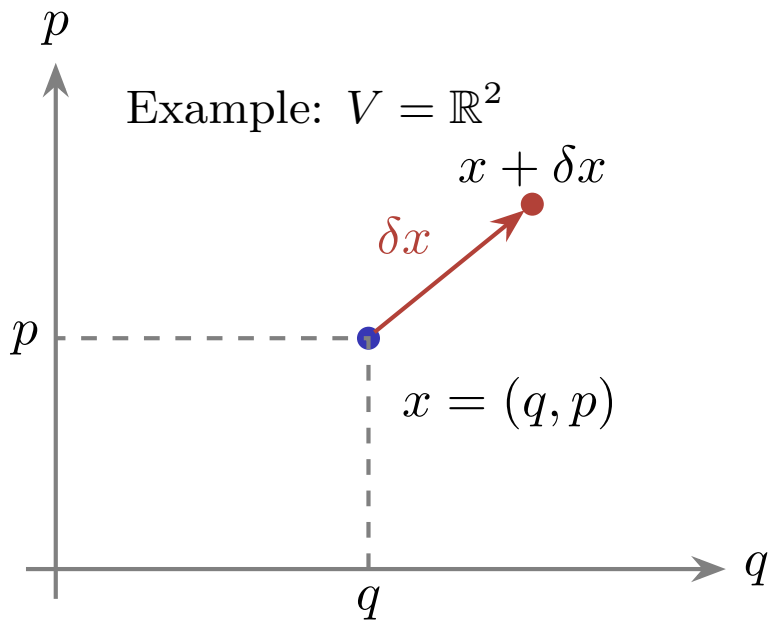
Endomorphism algebra

$$\begin{aligned} \mathrm{End}(V) &= \{A : V \to V : \ A \text{ linear}\} \\ &\cong M_d(\mathbb{R}). \end{aligned}$$

$$AB = A \circ B, \qquad I_V x = x.$$

Associative composition; unit I_V.

Euclidean geometry



$$\|x\| = \sqrt{x_1^2 + \cdots + x_d^2}.$$

The norm defines distance, continuity and differentiability.

Manuscript §1.1 and §1.2.1. Further algebra operations: Appendix A1.

From a velocity field to an ODE

A fluid element is carried by the local fluid velocity. Its position $x_t \in V$ evolves from an initial position x at time s according to

$$\begin{cases} \dot{x}_t = v_t(x_t), & t \in I, \\ x_s = x. \end{cases}$$

Here $I \subseteq \mathbb{R}$ is an open time interval, with $(s, x) \in I \times V$.

Velocity field

$$v : I \times V \longrightarrow V, \qquad (t, z) \longmapsto v_t(z),$$
$$v_t : V \longrightarrow V, \qquad z \longmapsto v_t(z).$$

At a fixed time t, v_t assigns a velocity to every position z.

Trajectory

$$x_\cdot : I \longrightarrow V, \qquad t \longmapsto x_t, \qquad x_\cdot \in C^1(I, V).$$

The solution traces one fluid element through time. Its tangent $\dot{x}_t$ equals the field evaluated at the current position, $v_t(x_t)$.

Manuscript §1.2.1, fluid-motion background and ODE initial-value problems.

Project 9.1: the initial-value problem

Given data

An open interval $I \subseteq \mathbb{R}$, with

$$A \in C(I, \mathrm{End}(V)),$$
$$B \in C(I, V).$$

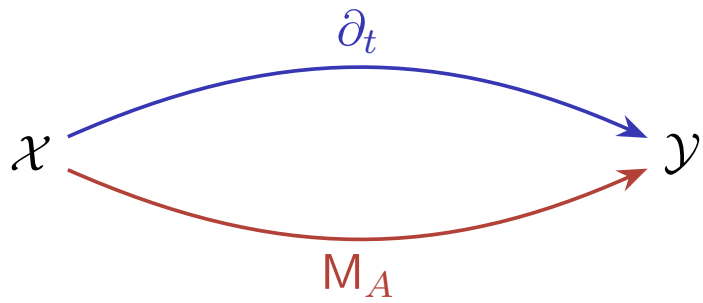
For $(s, x) \in I \times V$, solve

$$\begin{cases} \dot{x}_t = A_t x_t + B_t, & t \in I, \\ x_s = x. \end{cases}$$

Continuous coefficients give a unique trajectory on all of I.

$$\mathcal{X} = C^1(I, V), \qquad \mathcal{Y} = C(I, V),$$

Trajectory formulation



$$\mathsf{D}_A = \partial_t - \mathsf{M}_A$$

$$\mathsf{D}_A \gamma = B, \qquad E_s \gamma = x.$$

The unknown is a complete trajectory.

$$(\mathsf{M}_A \gamma)_t = A_t \gamma_t.$$

Assessment 9.1; Manuscript Example 1.10. Existence and uniqueness: Appendix A2.

Homogeneous evolution by evaluation

The homogeneous kernel

$$\mathcal{H} = \ker \mathsf{D}_A.$$

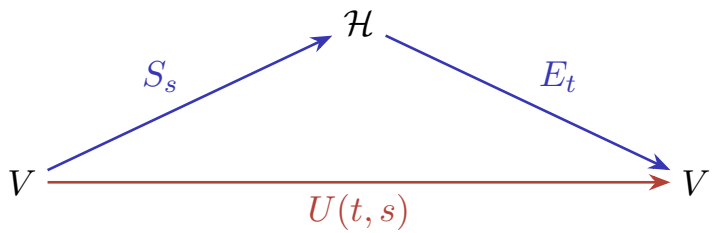
Evaluation identifies a trajectory with its state:

$$E_s : \mathcal{H} \to V, \qquad E_s(\eta) = \eta_s.$$

Existence and uniqueness make E_s a linear isomorphism.

$$S_s = E_s^{-1}, \qquad U(t,s) = E_t S_s.$$

Recover, then evaluate



$$\partial_t U(t,s) = A_t U(t,s),$$
$$U(s,s) = I_V.$$

Manuscript Example 1.10. Evaluation isomorphism and composition: Appendix A3.

The affine solution and its flow

Homogeneous and forced response

$$x_t = U(t,s)x + a_{s,t},$$

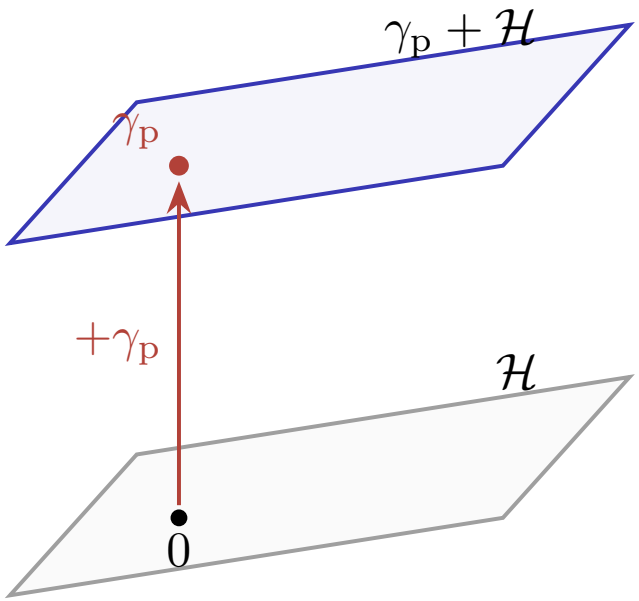
$$a_{s,t} = \int_s^t U(t,r)B_r \, \mathrm{d}r.$$

The initial value selects one trajectory from an affine family.

Initial-to-terminal state map

$$\Phi_{s,t}(x) = U(t,s)x + a_{s,t}.$$

For zero forcing, the map reduces to $U(t,s)x$.



Linear and affine subsets of $\mathcal{X}$

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

Manuscript Example 1.10. Affine family: A4; derivation: A5; verification: A6.

Project 9.2: general state flows

Allow a nonlinear velocity field

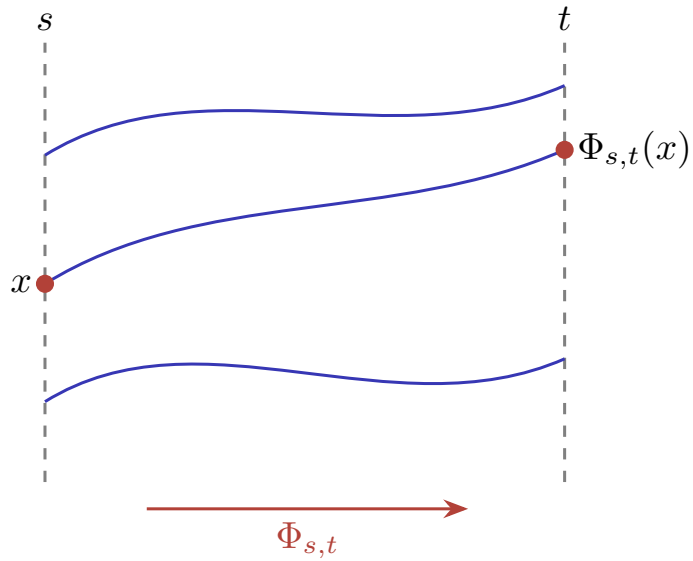
$$\dot{x}_t = v_t(x_t), \qquad x_s = x, \qquad \Phi_{s,t}(x) = x_t.$$

Assume continuous, locally Lipschitz dynamics with a unique trajectory on the same interval I for every initial pair.

The algebraic construction remains

$$\Phi_{s,t} = E_t \circ E_s^{-1}.$$

Evaluation is a bijection of sets; linearity of the state map is unnecessary.



Assessment 9.2; Manuscript Theorem 1.12. General evaluation construction: Appendix A7.

Composition, inverses, and reversal

Flow laws

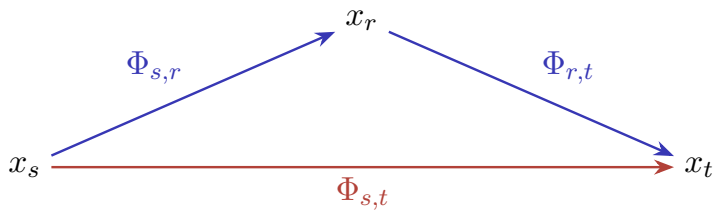
$$\Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t},$$
$$\Phi_{s,s} = I_V,$$
$$\Phi_{s,t}^{-1} = \Phi_{t,s}.$$

These identities hold for any $r, s, t \in I$.

Time reversal on a finite window

$$y_t = x_{T-t},$$
$$\dot{y}_t = -v_{T-t}(y_t), \qquad y_0 = x_T.$$

One trajectory, two routes



Uniqueness identifies the trajectory through the intermediate state.

Autonomy and completeness give a one-parameter group.

Manuscript Theorem 1.12. Composition: A7; autonomy: A8; time reversal: A9.

A nonlinear flow with an explicit solution

Triangular dynamics

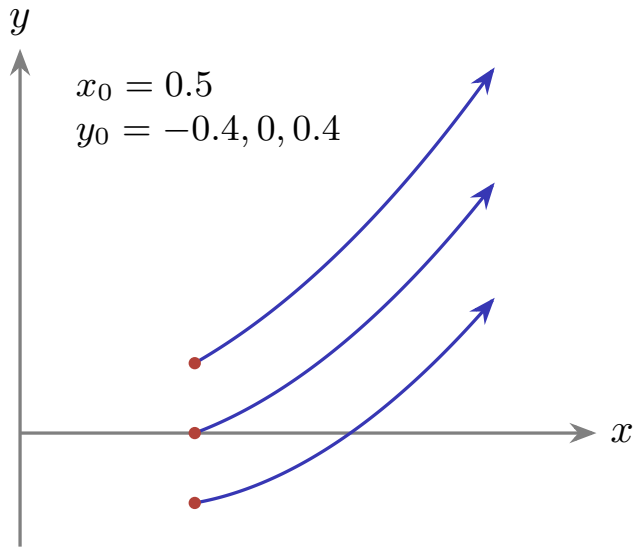
$$\dot{x}_t = 0.2x_t, \qquad \dot{y}_t = 0.1y_t + 0.3x_t^2.$$

Solve x first, then the forced equation for y.

Result for $\tau = t - s$

$$x_t = e^{0.2\tau}x,$$
$$y_t = e^{0.1\tau}y + x^2(e^{0.4\tau} - e^{0.1\tau}).$$

The flow is nonlinear in the state and exists for every real elapsed time.

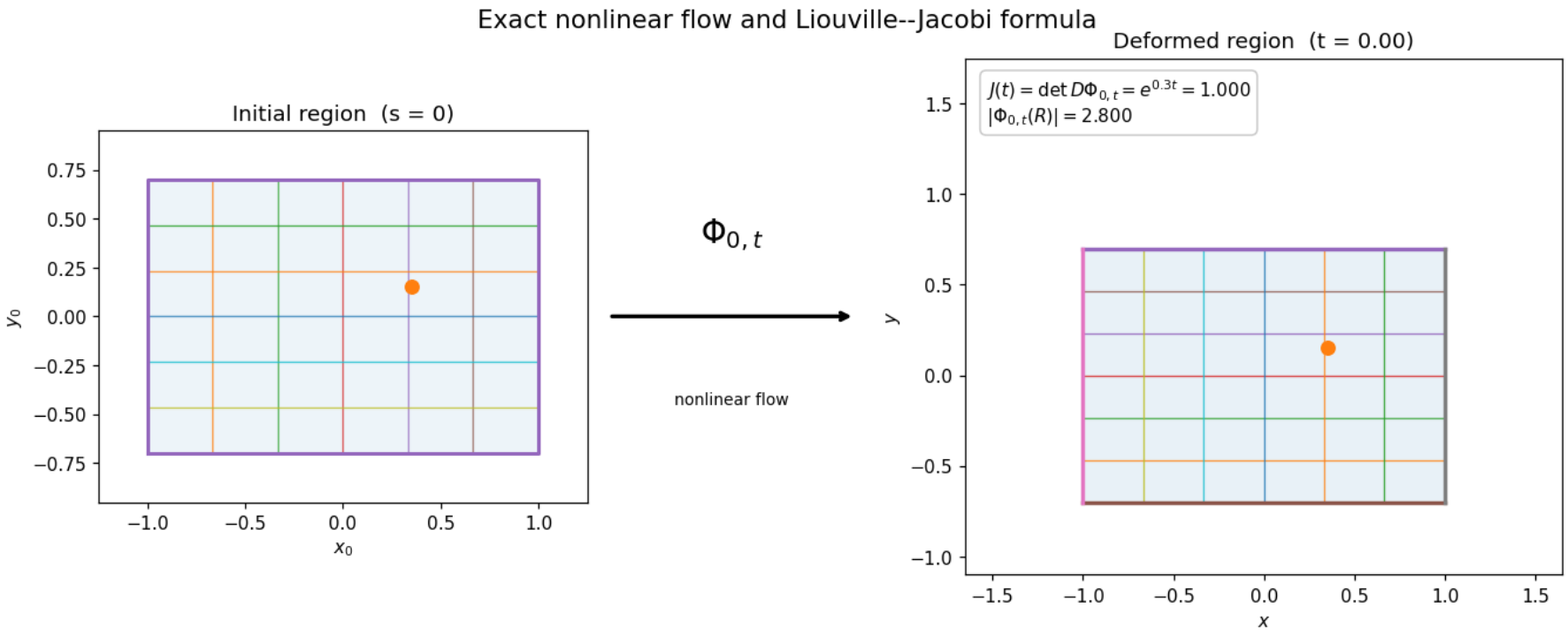


Trajectories for $0 \le t \le 5$

Manuscript Example 1.15. Curves evaluate the exact flow for three initial states.

A nonlinear flow in motion

$$R = [-1, 1] \times [-0.7, 0.7], \qquad 0 \le t \le 1.5.$$



The same flow moves every grid point. The quadratic coupling bends the rectangle.

Manuscript Example 1.15 and its accompanying 61-frame animation.

Evolution on the observable algebra

Pull back the observable

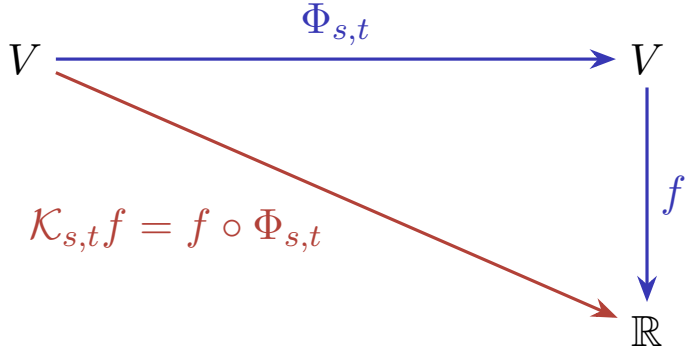
On the observable algebra $\mathcal{A} = B_b(V)$, define

$$\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}.$$

The state map may be nonlinear, while its induced action on functions is linear.

$$\mathcal{K}_{s,t}(fg) = (\mathcal{K}_{s,t} f)(\mathcal{K}_{s,t} g),$$
$$\mathcal{K}_{s,t} 1 = 1.$$

$$\mathcal{K}_{s,r}\mathcal{K}_{r,t} = \mathcal{K}_{s,t}.$$



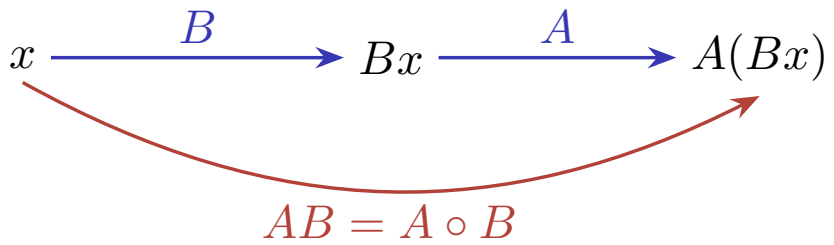
Manuscript §1.1, transformation and evolution operators. State maps are Borel measurable.

Appendix A1: algebra operations

The algebra $\mathrm{End}(V)$

$$(A+B)x = Ax + Bx,$$
$$(aA)x = a(Ax),$$
$$(AB)x = A(Bx).$$

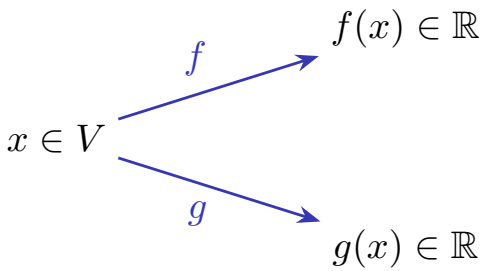
Composition is associative, with unit I_V; AB and BA can differ when $d \geq 2$.



The observable algebra $\mathcal{A} = B_b(V)$

$$(af + bg)(x) = af(x) + bg(x),$$
$$(fg)(x) = f(x)g(x),$$
$$1(x) = 1.$$

Bounded Borel functions form a commutative unital algebra.



Multiply values at the same state:
$(fg)(x) = f(x)g(x)$

Manuscript §1.1. Return to state space and algebra.

Appendix A2: existence and uniqueness

Bounds on a compact time window

For $v_t(y) = A_t y + B_t$ and compact $J \subset I$, let

$$L_J = \sup_{t \in J} \|A_t\|, \qquad C_J = \sup_{t \in J} \|B_t\|.$$

Both constants are finite, and

$$\|v_t(y) - v_t(z)\| \le L_J \|y - z\|,$$
$$\|v_t(y)\| \le C_J + L_J \|y\|.$$

Continuation argument

1. Local Lipschitz continuity gives a unique local solution.
2. Linear growth prevents finite-time escape on J.
3. Uniqueness joins the solutions on overlapping windows.

The trajectory is therefore uniquely defined on all of I.

Manuscript Example 1.10 and Appendix A.1. Return to the initial-value problem.

Appendix A3: evaluation and evolution

Homogeneous trajectories

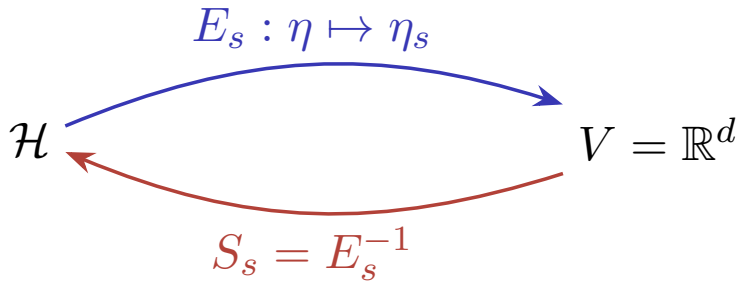
$$\mathcal{H} = \{\eta \in C^1(I, V) : \dot{\eta}_t = A_t \eta_t\}.$$

Evaluation $E_s(\eta) = \eta_s$ is linear.

- Existence makes it onto.
- Uniqueness makes it one-to-one.

$S_s = E_s^{-1}, \qquad \mathcal{H} \cong V, \qquad \dim \mathcal{H} = d.$

Inverse evaluation



$$E_s S_s = I_V, \qquad S_s E_s = I_{\mathcal{H}}.$$

Composition of the homogeneous evolution

$$U(t, r)U(r, s) = E_t S_r E_r S_s = E_t S_s = U(t, s).$$

Manuscript Example 1.10. Return to homogeneous evolution.

Appendix A4: the affine solution family

A particular solution shifts the kernel

Suppose $\mathsf{D}_A \gamma_{\mathrm{p}} = B$. Then

$$\begin{aligned} \mathsf{D}_A \gamma = B &\iff \mathsf{D}_A(\gamma - \gamma_{\mathrm{p}}) = 0 \\ &\iff \gamma - \gamma_{\mathrm{p}} \in \mathcal{H}. \end{aligned}$$

Hence

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

The initial condition selects one path

Write $\gamma = \gamma_{\mathrm{p}} + \eta$. Then

$$E_s \eta = x - \gamma_{\mathrm{p}}(s).$$

Since E_s is an isomorphism,

$$\gamma = \gamma_{\mathrm{p}} + S_s\big(x - \gamma_{\mathrm{p}}(s)\big).$$

Taking $\gamma_{\mathrm{p}}(s) = 0$ gives

$$\gamma = \gamma_{\mathrm{p}} + S_s x.$$

Manuscript Example 1.10. Return to the affine solution.

Appendix A5: variation of constants

1. Vary the homogeneous coefficient

$$\gamma_{\mathrm{p}}(t) = U(t,s)c(t), \qquad c(s) = 0.$$

2. Apply the trajectory operator

$$\begin{aligned} \dot{\gamma}_{\mathrm{p}} - A_t\gamma_{\mathrm{p}} &= U(t,s)\dot{c}(t), \\ \dot{c}(t) &= U(s,t)B_t. \end{aligned}$$

3. Integrate and compose

$$c(t) = \int_s^t U(s,r)B_r \,\mathrm{d}r.$$

Use $U(t,s)U(s,r) = U(t,r)$ to obtain

$$\begin{aligned} \gamma_{\mathrm{p}}(t) &= U(t,s)\int_s^t U(s,r)B_r \,\mathrm{d}r \\ &= \int_s^t U(t,r)B_r \,\mathrm{d}r. \end{aligned}$$

$$x_t = U(t,s)x + \int_s^t U(t,r)B_r \,\mathrm{d}r.$$

Manuscript Example 1.10. Return to the solution formula.

Appendix A6: verification of the solution

Initial condition

$$x_s = U(s,s)x + \int_s^s U(s,r)B_r \,\mathrm{d}r = x.$$

Differentiation

Since $\partial_t U(t,r) = A_t U(t,r)$, Leibniz's rule gives

$$\begin{aligned}
\dot{x}_t &= A_t U(t,s)x + U(t,t)B_t + \int_s^t A_t U(t,r)B_r \,\mathrm{d}r \\
&= A_t \left(U(t,s)x + \int_s^t U(t,r)B_r \,\mathrm{d}r \right) + B_t \\
&= A_t x_t + B_t.
\end{aligned}$$

Uniqueness

The difference of two solutions lies in $\mathcal{H}$ and has zero value at s. Injectivity of E_s makes that difference the zero trajectory.

Manuscript Example 1.10. Return to the solution and its flow map.

Appendix A7: general evaluation and composition

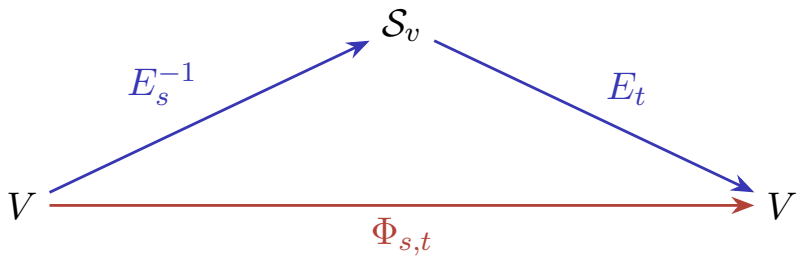
Evaluation is a bijection

$$\mathcal{S}_v = \{\gamma \in C^1(I, V) : \dot{\gamma}_t = v_t(\gamma_t)\}.$$

On the solution set, $E_s(\gamma) = \gamma_s$.
Existence gives surjectivity; uniqueness gives injectivity.

$$\Phi_{s,t} = E_t \circ E_s^{-1}.$$

Recover, then evaluate



Cancellation at an intermediate time

$$\begin{aligned}\Phi_{r,t} \circ \Phi_{s,r} &= (E_t \circ E_r^{-1}) \circ (E_r \circ E_s^{-1}) \\ &= E_t \circ E_s^{-1} = \Phi_{s,t}.\end{aligned}$$

$$\Phi_{s,s} = I_V, \qquad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

Manuscript Appendix A.2. Return to general flows; return to the flow laws.

Appendix A8: autonomy and completeness

Autonomous field

$$v_t = v, \qquad \Phi_{s,t} = \phi_{t-s}.$$

Uniqueness gives time-translation invariance. If all trajectories exist for every real time,

$$\begin{aligned} \phi_{\tau+\sigma} &= \phi_\tau \circ \phi_\sigma, \\ \phi_0 &= I_V, \\ \phi_{-\tau} &= \phi_\tau^{-1}. \end{aligned}$$

The maps form a one-parameter group.

Time-dependent field

$$v_t = v(t, \cdot).$$

Both initial and terminal times matter. The valid composition law is

$$\Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}.$$

The family $\Phi_{0,t}$ generally does not have the same one-parameter group law.

Manuscript Theorem 1.12 and Appendix A.2. Return to the main flow laws.

Appendix A9: time reversal

Reverse a finite time window

For a trajectory on $[0, T]$, set

$$y_t = x_{T-t}, \qquad 0 \le t \le T.$$

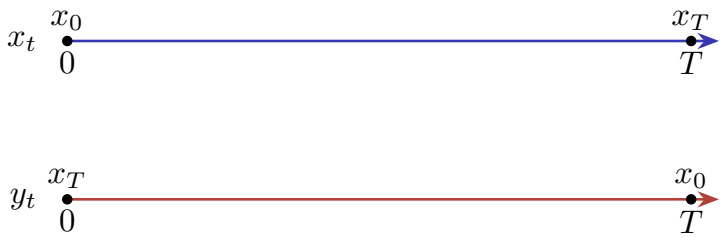
The chain rule gives

$$\begin{aligned} \dot{y}_t &= -\dot{x}_{T-t} \\ &= -v_{T-t}(x_{T-t}) \\ &= -v_{T-t}(y_t). \end{aligned}$$

Thus

$$\widetilde{v}_t = -v_{T-t}, \qquad y_0 = x_T.$$

Opposite traversal



Same state-space path, opposite traversal

Both the sign of the velocity and its time argument reverse.

Manuscript Theorem 1.12 and Appendix A.2. Return to composition and reversal.